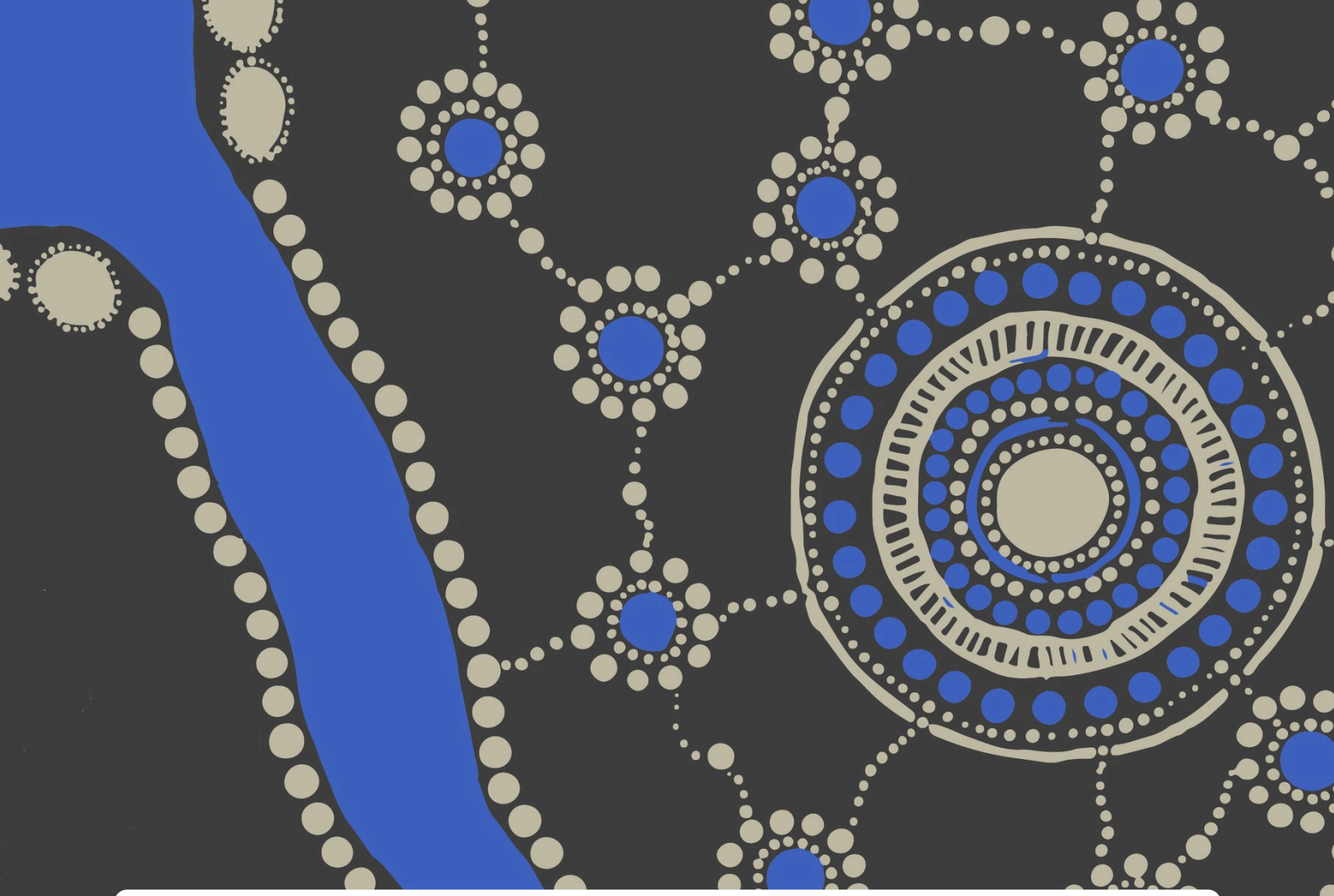


KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

Mathematics Standard 2
Year 12

CHAPTER 1 · KEY IDEAS AND REVIEW

Ratios and rates

Year 12 Standard 2 • Simplified review workbook

NSW Mathematics Standard 11–12 Syllabus (2024) • Outcomes MST-12-S2-05, MAO-WM-01

Name _____ Class _____ Teacher _____

Cover and banner artwork: *Kingscliff High School Vision Statement* by local Indigenous artist Marni Tuala.

About this workbook

This workbook is a simplified version of the **Chapter 1 key ideas and review** from your textbook, *CambridgeMATHS Mathematics Standard 2 Year 12*. There is one lesson for each key idea, and each lesson has three parts:

Key idea: the idea in plain words, with the *I can* statements from the textbook checklist.

Worked examples: every example is fully worked, with the reason for each line next to it. The answer is in a blue box.

Your turn: **multiple choice**, **short answer** and **extended response** questions, like the textbook review. Each question tells you where to look.

The workbook ends with a **chapter review** set out the same way, and answers to every question.

Contents

1	Ratios	1A	4
2	Capture–recapture	1B	7
3	Dividing a quantity in a given ratio	1C	10
4	Scale drawings	1D	13
5	Plans and elevations	1E	16
6	Perimeter and area of land	1F	19
7	Volume of water	1F	22
8	Rates	1G	25
9	The unitary method and best buys	1G	28
10	Speed and distance–time graphs	1G 1H	31
11	Energy	1I	34
12	Fuel consumption	1J	37
R	**Chapter review**		40
A	Answers		45

WORKING ON YOUR OWN?

1. Read the key idea and look at the picture.
2. Read a worked example one line at a time. Cover the next line: can you say what comes next?
3. Try the questions that point to that example. Write every step in the box.
4. Check the answers at the back, fix any mistakes, then rate yourself on the checklist.

Syllabus outcomes

MST-12-S2-05	solves practical problems involving ratios and rates
MAO-WM-01	develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly

Formulas you'll use (some are on the HSC reference sheet)

Trapezoidal rule: $A \approx \frac{h}{2}(d_f + d_l)$

Speed: $S = \frac{D}{T}$

Volume: $V = Ah$, $1\ \text{m}^3 = 1000\ \text{L}$

Fuel used (L) $= \frac{\text{distance (km)}}{100} \times \text{rate (L/100 km)}$

Energy (kWh) = power (kW) × time (h)

Acknowledgement of Country

We acknowledge the Traditional Custodians of the land on which we learn, the Bundjalung people, and pay our respects to Elders past and present. The artwork on the cover and throughout this workbook is our school vision statement, created by local Indigenous artist Marni Tuala.

You will need

A calculator, a ruler and a pencil. Show your working in every box: in the HSC, working earns marks even when the final answer is wrong.

Key ideas Chapter 1 at a glance

A simplified version of the textbook's *Key ideas and chapter summary*. Each key idea has its own lesson.

			EXAMPLE
1	Ratio	Compares amounts of the **same kind**, in a set order. Multiply or divide every part by the same number to get an **equivalent ratio**.	5 : 2 is not the same as 2 : 5 5 : 2 = 10 : 4 = 15 : 6
2	Capture–recapture	Estimates the size of an animal population. Tag some animals, release them, then catch a second sample.	$\frac{\text{tagged in 2nd sample}}{\text{size of 2nd sample}} = \frac{\text{tagged at first}}{\text{population}}$
3	Dividing a quantity in a ratio	**1** Add the parts (2 + 3 = 5). **2** Write each part over the total. **3** Multiply the quantity by each fraction.	Share \$40 in the ratio 2 : 3. $\frac{2}{5} \times \$40 = \$16$, $\frac{3}{5} \times \$40 = \$24$
4	Scale drawings	Scale = drawing length : actual length. A scale can have units (1 cm to 1 m) or no units (1 : 100).	Scale 1 : 100. 3 cm on the plan → real length = 3 × 100 = 300 cm = 3 m
5	Plans and elevations	A **plan** is the view from the top. An **elevation** is the view from one side, such as the front or the side.	plan = bird's-eye view front elevation = view from the front
6	Perimeter and area of land	**Perimeter**: measure each side on the photo and multiply by the scale. **Area**: count grid squares to estimate, or use a formula. For one curved side, use the trapezoidal rule.	$A \approx \frac{h}{2}(d_f + d_l)$ h = distance between the two measurements, d_f = first, d_l = last
7	Volume of water	If the depth is the same everywhere, volume = area of the surface × depth. For rain, change millimetres to metres first.	$V = Ah$ (h = depth), 1 m^3 = 1000 L 18 mm = 0.018 m
8	Rates	Compares two amounts with **different units**.	km/h, \$/kg, L/100 km, \$/hour 25 mL/min = 25 × 60 mL/h = 1500 mL/h = 1.5 L/h
9	Unitary method	Divide to find the amount for **1**. Multiply to find the amount you need. For the best buy, compare prices for the same amount.	3 kg costs \$12 1 kg: \$12 ÷ 3 = \$4 → 5 kg: 5 × \$4 = \$20
10	Speed	Speed is a rate. On a distance–time graph, a steeper line is faster and a flat line means stopped.	$S = \frac{D}{T}$, $D = S \times T$
11	Energy	Energy (kWh) = power (kW) × time (hours). Running cost = energy × price per kWh.	1 kW = 1000 W, so 600 W = 0.6 kW 2 kW for 3 h = 6 kWh at \$0.30 per kWh: 6 × \$0.30 = \$1.80
12	Fuel consumption	A rate given in litres per 100 km (L/100 km) or km per litre.	Fuel used (L) = $\frac{\text{distance (km)}}{100} \times \text{rate}$ Rate = $\frac{\text{litres}}{\text{km}} \times 100$ 30 L for 400 km → 7.5 L/100 km

1 Ratios

Key idea: Ratio • Textbook 1A • MST-12-S2-05

Key idea Ratio

A **ratio** compares two or more amounts of the **same kind**, in a set order.



sand : cement = 5 : 2

- 5 buckets of sand to 2 buckets of cement is 5 : 2, read "5 to 2".
- **Order matters:** cement to sand is 2 : 5. A ratio has **no units**.
- **Equivalent ratios:** multiply or divide **every** part by the same number. 5 : 2 = 10 : 4 = 15 : 6

Simplest form

Divide every part by the **highest common factor** (HCF).

- **Finding the HCF:** try the factors of the smallest number, biggest first, until one divides into every part.
- **Decimals?** Multiply every part by 10 first.
- **Different units?** Make them the same: 40 cm : 2 m = 40 : 200 = 1 : 5.

Three ways to write a ratio

Form	How	8 : 10
Simplest	÷ by the HCF	4 : 5
1 : a	÷ by the **first** number	1 : 1.25
b : 1	÷ by the **second** number	0.8 : 1

Finding an unknown

Find two numbers in the same place. Divide to find what one was multiplied by. Do the **same** to the other part.

$3 : 4 = a : 20 \rightarrow 20 \div 4 = \mathbf{5}$, so $a = 3 \times \mathbf{5} = 15$

CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* change a ratio to simplest form, to the form 1 : a and to the form b : 1. ○○○
- ▸ *I can* use a ratio to find an unknown amount. ○○○

WORKED EXAMPLE 1.1 Simplest form

Write each ratio in simplest form. (a) 12 : 18 (b) 10 : 15 : 30 (c) 4.8 : 1.6

	WORKING	WHY
(a)	12 : 18	The HCF of 12 and 18 is **6**.
	= 12 ÷ 6 : 18 ÷ 6	Divide **both** parts by 6.
	= 2 : 3	Only 1 divides into 2 and 3, so this is simplest form.
(b)	10 : 15 : 30	The HCF of 10, 15 and 30 is **5**.
	= 10 ÷ 5 : 15 ÷ 5 : 30 ÷ 5	Divide **every** part by 5.
	= 2 : 3 : 6	Only 1 divides into 2, 3 and 6.
(c)	4.8 : 1.6	There are decimals. Make whole numbers first.
	= 4.8 × 10 : 1.6 × 10	Multiply **both** parts by **10**.
	= 48 : 16	The HCF of 48 and 16 is **16**.
	= 48 ÷ 16 : 16 ÷ 16	Divide both parts by 16.
	= 3 : 1	Keep the 1. Write 3 : 1, not just 3.

WORKED EXAMPLE 1.2 — The forms 1 : a and b : 1

Write the ratio 15 : 12 (a) in the form $1 : a$ (b) in the form $b : 1$.

(a) $15 : 12$ — To make the **first** part 1, divide both parts by the first number, 15.

$= 15 \div 15 : 12 \div 15$ — Use your calculator: 12 ÷ 15 = 0.8

$= \boxed{1 : 0.8}$ — The first part is now 1.

(b) $15 : 12$ — To make the **second** part 1, divide both parts by the second number, 12.

$= 15 \div 12 : 12 \div 12$ — Use your calculator: 15 ÷ 12 = 1.25

$= \boxed{1.25 : 1}$ — The second part is now 1.

WORKED EXAMPLE 1.3 — Finding an unknown

Find the value of the pronumeral. (a) $2 : 5 = a : 20$ (b) $3 : 4 = 15 : b$

(a) $2 : 5 = a : 20$ — Find two numbers in the same place: 5 and 20 are both **second** parts.

$20 \div 5 = \mathbf{4}$ — Divide to find what 5 was multiplied by ($5 \times 4 = 20$).

$a = 2 \times \mathbf{4}$ — Do the same to the first part.

$\boxed{a = 8}$ — Check: 8 : 20 simplifies to 2 : 5.

(b) $3 : 4 = 15 : b$ — 3 and 15 are both **first** parts.

$15 \div 3 = \mathbf{5}$ — Divide to find what 3 was multiplied by ($3 \times 5 = 15$).

$b = 4 \times \mathbf{5}$ — Do the same to the second part.

$\boxed{b = 20}$ — Check: 15 : 20 simplifies to 3 : 4.

WORKED EXAMPLE 1.4 — Using a ratio to find an amount

The ratio of boys to girls in a class is 3 : 4. There are 12 girls. How many boys are in the class?

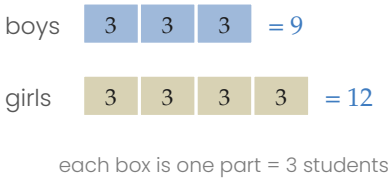
boys : girls = 3 : 4 — Write the ratio with labels so you know which part is which.

4 parts = 12 girls — Girls are the **second** part, so 4 parts = 12.

1 part = 12 ÷ 4 = 3 — Find **one part** first.

boys = 3 parts = 3 × 3 — Boys are the **first** part, 3 parts.

= 9 boys — Check: 9 : 12 simplifies to 3 : 4.



YOUR TURN 1

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 Which is equivalent to 2 : 5? → Key idea

A. 4 : 10 **B.** 5 : 2
C. 4 : 7 **D.** 2 : 10

2 What is 15 : 20 in simplest form? → Example 1.1

A. 5 : 4 **B.** 3 : 4
C. 1 : 5 **D.** 30 : 40

3 Adults : children = 5 : 9. There are 630 children. How many adults?
Hint: 9 parts = 630. → Example 1.4

A. 70 **B.** 126
C. 280 **D.** 350

Short answer show your working in the box

1 Write each ratio in simplest form. *Hint: find the HCF first.* → Example 1.1

(a) 20 : 30 (b) 6 : 9 : 12 (c) 1.5 : 2.5

(a)	(b)	(c)

2 Consider the ratio 10 : 4. Write it → Examples 1.1 and 1.2

(a) in simplest form (b) in the form 1 : a (c) in the form b : 1

(a)	(b)	(c)

3 Find the value of the pronumeral. *Hint: divide to find what was multiplied.* → Example 1.3

(a) 2 : 3 = a : 15 (b) 4 : 3 = a : 24 (c) 3 : 5 = 12 : b

(a)	(b)	(c)

4 Gabriel mixes sand and cement in the ratio 5 : 2. He uses 4 buckets of cement. How many buckets of sand should he use?
Hint: 2 parts = 4 buckets. → Example 1.4

5 The ratio of Victoria's height to Willow's height is 8 : 7. Victoria is 176 cm tall. How tall is Willow?
Hint: 8 parts = 176 cm. → Example 1.4

Extended response answer each part in order

1 A fruit punch recipe mixes orange juice and lemonade in the ratio 3 : 5. → Key idea and Example 1.4

(a) Write the ratio of **lemonade** to **orange juice**.

(b) Mia uses 600 mL of orange juice. How many millilitres is **one part**?

(c) How much lemonade does Mia need?

(d) How much punch does Mia make altogether? *Hint: add the orange juice and the lemonade.*

(a)	(b)	(c)	(d)

Answers

Lesson 1 Ratios

MULTIPLE CHOICE

1 A **2** B

3 D (1 part = 630 ÷ 9 = 70, adults = 5 × 70 = 350)

SHORT ANSWER

1 (a) 2 : 3 (b) 2 : 3 : 4 (c) 3 : 5

2 (a) 5 : 2 (b) 1 : 0.4 (c) 2.5 : 1

3 (a) $a = 10$ (b) $a = 32$ (c) $b = 20$

4 1 part = 4 ÷ 2 = 2,
sand = 5 × 2 = 10 buckets

5 1 part = 176 ÷ 8 = 22 cm,
Willow = 7 × 22 = 154 cm

EXTENDED RESPONSE

1 (a) 5 : 3
(b) 600 ÷ 3 = 200 mL
(c) 5 × 200 = 1000 mL
(d) 600 + 1000 = 1600 mL